Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/15/2007
PI Name:	Sonnenfeld, Gerald Ph.D.		
Project Title:	Bed Rest and Immunity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Binghamton University, State University of New York		
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PI Web Page:			
City:	Binghamton	**State:**	NY
Zip Code:	13902-6000	**Congressional District:**	5
Comments:	PI moved to Binghamton University, from Morehouse School of Medicine, in 2004.		
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-05
Start Date:	07/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is 6/30/2007, per K. Major/NSBRI (12/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Butel, Janet (Baylor College of Medicine) Shearer, William (Baylor College of Medicine)		
Grant/Contract No.:	NCC 9-58-IIH00406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exposure of humans to space flight conditions results in changes in immune responses. The contribution of these changes in the immune system to alterations in resistance to infection and tumors remains to be fully established. Chronic bed-rest with a head-down tilt has been a human model of choice for the effects of space flight on physiological systems. This model allows for no load-bearing on the legs of the subjects as well as a fluid shift to the head. Results utilizing the head-down tilt bed-rest model to study the effects of space flight conditions on the immune system are limited, but show a pattern similar to the results observed when humans are exposed to space. The head-down tilt bed-rest model has been established as valid for studying the effects of space flight conditions on functional dynamic immune responses such as cytokine production, but all of the studies to date have been carried out on men. The proposed study is designed to determine the effects of bed-rest on immune responses of women. The overall hypothesis to be tested is: maintenance of female subjects in the head-down tilt bed-rest model will result in suppression of functional immune responses and enhance susceptibility to infection. The specific aims of the proposed study are: to determine effects of maintenance of female subjects in the head-down tilt bed-rest model on functional immune responses. and to determine effects of maintenance of female subjects in the head-down tilt bed-rest model on resistance to infection. By carrying out these studies, we will correlate the effects of bed rest on immune function and control of viral infections. The French bed rest campaigns have been completed and the results are under anaylsis at the present time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many elderly individuals must remain in chronic bed rest and often develop decubiti (infected bed sores). The results of this study may aid in understanding mechanisms and in providing treatment for these patients. The analysis of the results of the study will provide new insight in this area as well as in the validity of the bed rest model for determining the effects of space flight conditions and countermeasures on immune responses of women.
Task Progress:	Preliminary data have been obtained on viral reactivation and on production of antibody against phiX-174. The bed rest study is complete, but additional time will be required for all of the data to be analyzed and for the manuscripts generated to be prepared.
Bibliography Type:	Description: (Last Updated: 04/28/2011)
Abstracts for Journals and Proceedings	Sonnenfeld G. "Cytokines and interferons in the spaceflight environment." October 2005. Cell Res. 2005 Oct;15:103. , Oct-2005
Articles in Peer-reviewed Journals	Sonnenfeld G, Aviles H, Butel JS, Shearer WT, Niesel D, Pandya U, Allen C, Ochs HD, Blancher A, Abbal M. "Bed rest and immunity." Acta Astronaut. 2007 Feb-Apr;60(4-7):234-6. http://dx.doi.org/10.1016/j.actaastro.2006.08.004 , Feb-2007
Articles in Peer-reviewed Journals	Sonnenfeld G. "Exploration class missions and return: effects on the immune system." Gravit Space Biol. 2006 Aug;19(2):45-7. , Aug-2006
Articles in Peer-reviewed Journals	Sonnenfeld G. "The immune system in space, including Earth-based benefits of space-based research." Curr Pharm Biotechnol. 2005 Aug;6(4):343-9. Review. PMID: 16101473 , Aug-2005
Articles in Peer-reviewed Journals	Sonnenfeld G. "Experimentation with animal models in space. Introduction." Adv Space Biol Med. 2005;10:1-5. PMID: 16101102 , Jun-2005
Articles in Peer-reviewed Journals	Sonnenfeld G. "Use of animal models for space flight physiology studies, with special focus on the immune system." Gravit Space Biol Bull. 2005 Jun;18(2):31-5. Review. PMID: 16038091 , Jun-2005